

Xinming Tu

PhD Candidate, University of Washington (expected 2026)

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RESEARCH INTERESTS

My current research focuses on AI systems for autonomous scientific reasoning and discovery, including LLM-based agentic architectures, rigorous benchmarks for AI evaluation, and multi-modal foundation models. I also work on AI for regulatory genomics, developing methods for sequence-to-function prediction and multi-omics integration.

SELECTED PUBLICATIONS

*Equal contribution, †Corresponding author

- **CLUE**: Xinming Tu*, Zhi-jie Cao*, Chen-rui Xia, Sara Mostafavi†, Ge Gao†. *Cross-Linked Unified Embedding for cross-modality representation learning*, **NeurIPS 2022**, **Oral Presentation (top 1.7%)**
- **DeepAllele**: Xinming Tu, Sara Mostafavi. *Deciphering Allele-Specific Regulation Across Chromatin Accessibility, Protein Binding, and Gene Expression using DeepAllele*, **RECOMB 2026**
- **SAGE-net**: Anna E Spiro*, Xinming Tu*, Yilun Sheng, Alexander Sasse, Rezwan Hosseini, Maria Chikina, Sara Mostafavi. *A scalable approach to investigating sequence-to-expression prediction from personal genomes*, **Nature Methods 2026** (Accept in Principle)
- **LLM4DS**: Xinming Tu, James Zou, Weijie J. Su†, Linjun Zhang†. *What Should Data Science Education Do with Large Language Models?*, **Harvard Data Science Review 2023**
- **SC-VAE**: Xinming Tu, Jan-Christian Hutter, Zitong Jerry Wang, Takamasa Kudo, Aviv Regev, Romain Lopez. *A Supervised Contrastive Framework for Learning Disentangled Representations of Cell Perturbation Data*, **MLCB 2023**

Other Publications: SLAT (**Nature Communications 2023**), vCNN (**Briefings in Bioinformatics 2021**), ePooling (**Bioinformatics 2020**)

AI RESEARCH PROJECTS

- **University of Washington** Seattle, WA
PhD student advised by Sara Mostafavi Sep 2021 - Present
 - **Paper2Bench: Multi-Agent System for Automated Benchmark Generation**: *In Preparation*
 - Designed a multi-agent LLM system that automatically converts scientific papers into executable benchmark tasks with verified ground truth, enabling scalable AI evaluation.
 - Developed a 4-agent pipeline (Extractor, Planner, Validator, Solver) with iterative refinement loops for feasibility checking, task generation, and solution synthesis.
 - Built sandboxed execution environment with Docker for reproducible evaluation; supports multi-modal scoring across code, numerical, and textual outputs.
 - **GrantBench: Benchmarking LLM Scientific Reasoning and Planning**: *In Preparation*
 - Developed a progressive two-track benchmark to evaluate AI systems on scientific reasoning, from constrained research planning (Track 1) to open-ended proposal ideation (Track 2).
 - Designed rigorous evaluation framework based on expert grant review criteria (Significance, Innovation, Feasibility, Clarity) with longitudinal blind hold-out sets using latest funded grants to prevent overfitting.
 - Built automated parsing pipeline to extract structured proposal components (background, significance, aims, methodology) from NIH and NSF grants for systematic assessment.

COMPUTATIONAL BIOLOGY RESEARCH

- **University of Washington** Seattle, WA
PhD student advised by Sara Mostafavi Sep 2021 - Present
 - **DeepAllele: Interpretable Deep Learning for Causal Variant Prediction**: **RECOMB 2026**
 - Developed and implemented DeepAllele, an interpretable deep learning architecture to predict allele-specific regulatory activities from sequence data in ATAC-seq, ChIP-seq, and RNA-seq experiments.
 - Identifies functional genetic variants driving allelic imbalances across chromatin accessibility, TF binding, and gene expression.
 - **DRUM: Sequence Model for Multi-layer Gene Regulation**:
 - Developed a sequence-based deep neural network for constructing a multi-layered view of gene expression, chromatin accessibility, and regulatory sequences.
 - Models both *cis* and *trans* regulatory processes with multi-task learning across cell types.
 - **CLUE: Cross-Linked Unified Embedding for Multi-modal Learning**: **NeurIPS 2022 Oral**
 - Proposed a semi-supervised neural network using cross-linked embeddings for multi-modality representation learning.
 - Achieved state-of-the-art performance integrating gene expression, protein abundance, and DNA accessibility data.

EDUCATION

- **University of Washington**
PhD in Computer Science & Computational Molecular Biology
Advisor: Sara Mostafavi
 - **Peking University**
B.S. in Biological Science & Computer Science
Undergraduate Honors Program in Biology
- Seattle, USA
Sep 2021 - Present

Beijing, China
Sep 2016 - June 2021

INDUSTRY EXPERIENCE

- **Genentech R&D**
Research Intern, Aviv Regev Lab
 - **Microsoft Research Asia**
Research Intern, Machine Learning and Computational Biology Group
- San Francisco, USA
June 2023 - Sep 2023

Beijing, China
Oct 2019 - Apr 2020

TECHNICAL SKILLS

- **Deep Learning:** PyTorch, **JAX/Flax**, Transformers, Distributed training, Mixed-precision training
- **LLM Systems:** Multi-agent architectures, Prompt engineering, RAG, LLM-as-judge evaluation
- **Infrastructure:** Docker, Slurm (HPC), GCP, Large-scale data pipelines, Weights & Biases
- **Languages:** Python (expert), R, C/C++, Shell, $\text{\LaTeX}$

ACADEMIC SERVICE

- **Conference Reviewer:** **NeurIPS** (2022-2025), **ICML** (2024-2025), **ICLR** (2024-2025), AISTATS (2024), AAAI (2024), KDD (2023), RECOMB (2021-2022), MLCB (2022), ISMB/ECCB (2023)
- **Journal Reviewer:** PLOS Computational Biology
- **Admission Reviewer:** UW CSE Ph.D. Admission (2021-2022)

SELECTED AWARDS

- Wissner-Slivka Endowed Fellowship in Computer Science & Engineering, UW
 - Paul G. Allen First-Year Graduate Student Fellowship
 - ShenTong Graduates (Highest Graduate Honor, School of Life Sciences, PKU)
 - Outstanding Graduates of Peking University
 - Outstanding Graduates of Beijing
 - National Olympiad in Informatics in Provinces (NOIP) First Prize
- 2025-2026

2021

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2014